AQUAFAB INDUSTRIES

Running a Respirometry Trial

Intermittent-flow and closed respirometry, from setting up to saving the result.

For Aquafab Workbench 0.2.0

Edition of 28 October 2026

Screenshots show simulated machines and modelled example data, not measurements. Aquafab Workbench is not a safety system.

aquafabindustries.com · sales@aquafabindustries.com · +1 (513) 781-8010

Contents

1 **Running a respirometry trial**

- 1.1 How intermittent-flow respirometry works
- 1.2 Set up
- 1.3 Prepare the machine
- 1.4 Run
- 1.5 Save
- 1.6 Other measurements on the same page
- 1.7 Good practice

1 Running a respirometry trial

This guide runs an intermittent-flow respirometry trial, the standard way to measure standard metabolic rate (SMR), from setting it up to saving the result. The same page also runs MMR, aerobic scope, P_{crit} and EPOC measurements; how they differ is at the end.

You need: a machine with a dissolved-oxygen probe, the probe calibrated (*Commissioning a New Machine*, step 5), the chamber's volume entered in **Settings → This machine's limits**, and a license to start a trial on connected hardware. Rehearse on a simulated machine first; it needs no license.

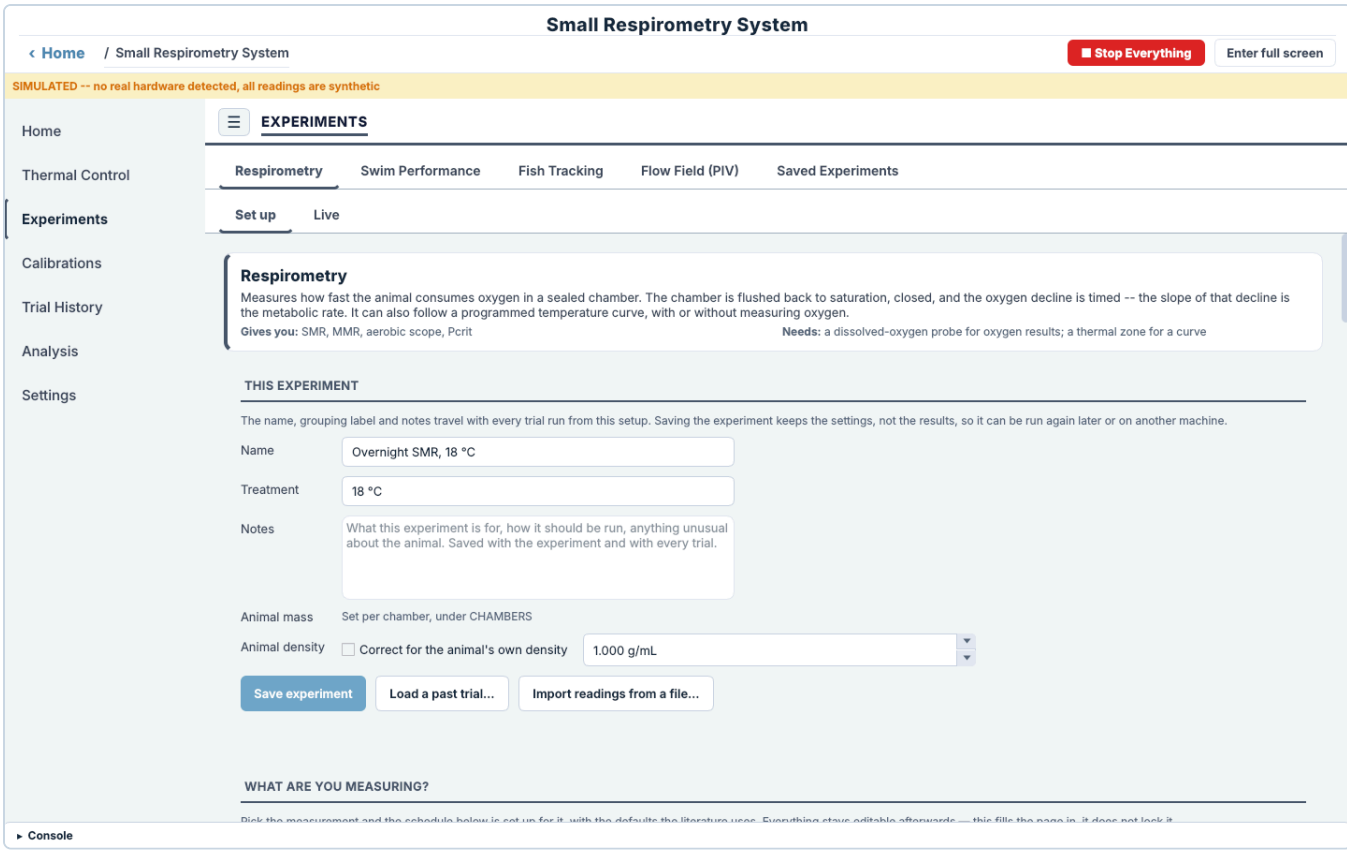
> Screenshots are from a simulated machine. Numbers in them are not measurements.

1.1 How intermittent-flow respirometry works

The chamber is **flushed** with fresh water back to saturation, left to **wait** a moment while the water mixes, then **closed** while the animal's oxygen uptake draws the oxygen down. The slope of that decline, times the water volume, divided by the animal's mass, is one MO_2 (mg O_2 kg^{-1} h^{-1}). Repeat for many cycles, and SMR is estimated from the low end of them (Chabot, Steffensen & Farrell 2016).

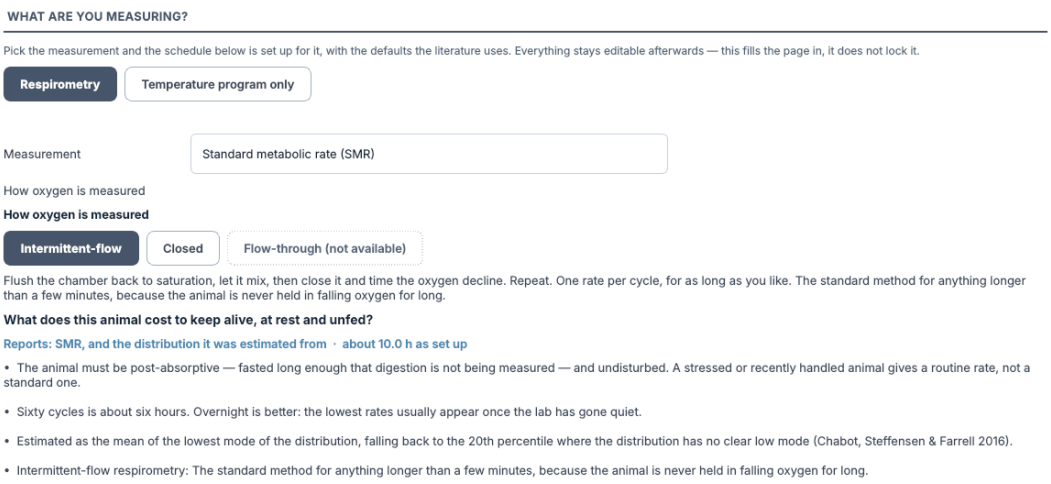
1.2 Set up

Open the machine, then **Experiments → Respirometry → Set up**. Work down the cards.



The top of the Set up view.

1.2.1 What are you measuring?

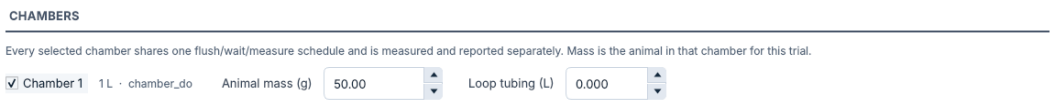


1. Choose **Respirometry** (not *Temperature program only*).
2. **Measurement: Standard metabolic rate (SMR)**. This fills in the literature defaults: 1 h acclimation, 60 cycles of 180 s flush, 60 s wait, 300 s measure, minimum R^2 0.95, SMR by MLND. About 10 hours. Everything stays editable.
3. **How oxygen is measured: Intermittent-flow.**

1.2.2 This experiment

- **Name**, for example *SMR overnight 18 °C*.
- **Treatment**, for example *18C*. Trials sharing a treatment are compared together in Analysis, so type it identically for every animal in a group.
- **Notes**: anything a reader would want to know.

1.2.3 Chambers



Tick each chamber in use and enter, for each:

- **Animal mass (g)**: this animal, weighed today.
- **Loop tubing (L)**: water in the chamber's recirculation loop, if any.

Read the line under the fields, for example *"0.953 L of water: 1.000 L chamber − 47 mL displaced by the animal"*. That is the volume every rate is multiplied by. If it looks wrong, the chamber volume in Settings is wrong.

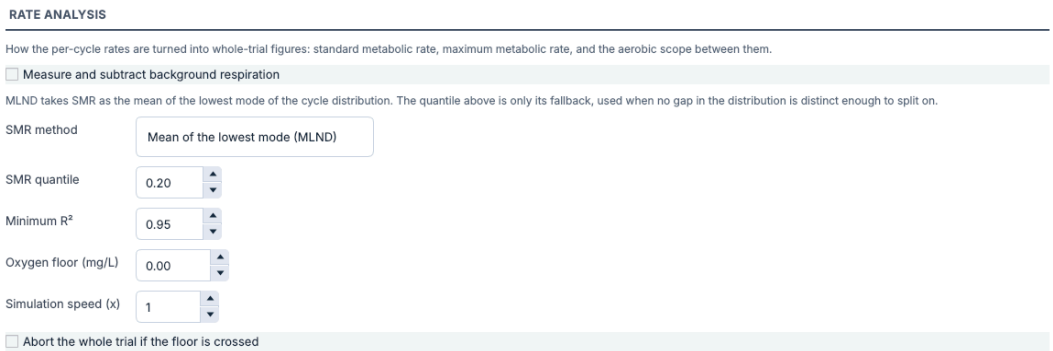
1.2.4 Acclimation, measurement cycle, water speed

- **Acclimation**: how long the animal settles before measuring, flushed to keep it oxygenated.
- **Measurement cycle**: **Cycles**, **Flush (s)**, **Wait (s)**, **Measure (s)**. Size the measure phase so oxygen falls usefully but not too far: a big animal in a small chamber needs a shorter measure. **Total protocol duration** adds it up.
- **Water speed**: **Leave as set**, or **Constant** to record the speed with the trial.

1.2.5 Temperature

Set **Constant temperature**. Leave the curve button at **OFF — holding one temperature** for a held-temperature SMR.

1.2.6 Rate analysis



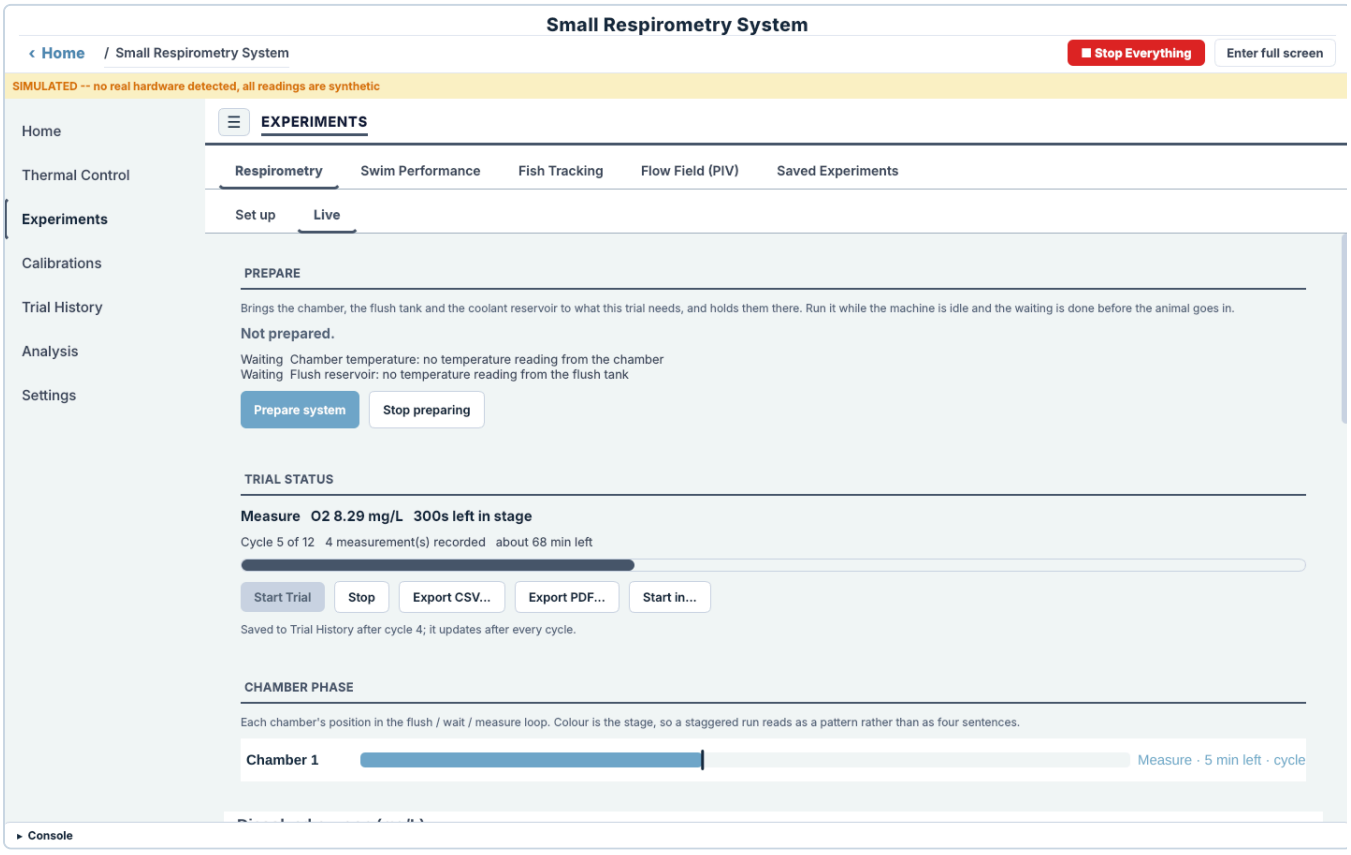
- **Measure and subtract background respiration**: adds a blank cycle and subtracts its rate (Svendsen et al. 2016). Recommended.
- **SMR method**: *Low quantile* or *MLND*; **SMR quantile** 0.2.
- **Minimum R^2**: cycles fitting worse are kept but excluded.
- **Oxygen floor (mg/L)**: if oxygen falls below this in a closed phase, the chamber is flushed at once. **Set it.**

Press **Save experiment** to keep the whole setup for the next animal.

1.3 Prepare the machine

Switch to **Live** and press **Prepare system**, before the animal goes in. The Prepare card lists what it is waiting for: the chamber at temperature, the flush tank at the same temperature (so a flush is not a thermal shock), the reservoir chilled. Wait until every line is ready.

1.4 Run



The Live view during a simulated trial.

1. Put the animal in, close the chamber, and check for bubbles.
2. Press **Start Trial** (**Ctrl+Return**). Or **Start in…** with a delay in minutes, to start after you have left the room.

3. Watch the first cycles. The **Dissolved oxygen** chart should fall steadily in each measure phase and recover with each flush; each cycle adds a point to **MO_2**. **Chamber phase** shows each chamber's stage.
4. **Derived statistics** update as cycles complete: SMR, MMR, aerobic scope, mean R^2.

If anything goes wrong, **Stop** ends the trial (completed cycles are kept), and ■ **Stop Everything** stops the whole machine.

1.5 Save

There is nothing to press: the trial saves itself to Trial History after the first cycle and again after every cycle, so a power cut at six in the morning keeps the night's cycles. The line under the buttons says how far it has got, and reads *"Trial saved to Trial History (N cycle(s))."* when the run ends. A run that never finished is marked *Interrupted* in Trial History. Optionally, **Export CSV…** (every cycle and the oxygen trace) or **Export PDF…** (a printed report).

The trial is now in **Trial History**, where **Inspect** shows every cycle and every setting used.

1.6 Other measurements on the same page

Measurement	What changes
Routine metabolic rate (RMR)	Shorter run; reports the mean of accepted cycles.
Maximum metabolic rate (MMR), after a chase	No acclimation, short cycles. Chase the animal to exhaustion, move it in, and start **immediately**.
Aerobic scope	SMR and MMR in one run, with absolute and factorial scope.
Critical oxygen level (P_{crit})	**Closed** respirometry: one long closed phase while oxygen falls. **Set the oxygen floor.** P_{crit} appears in Derived statistics once the oxygen range allows a fit.
Recovery after exercise (EPOC)	No acclimation, many short cycles following recovery.

1.7 Good practice

- Fast the animal as your protocol requires, and run overnight where you can.
- Check the respirometric volume line before every trial.
- In **Analysis → Work on a trial**, you can leave out the first cycles while the animal was settling, without changing the saved trial (*Analysing and Exporting Results*).